

ID3 Algorithm

- هذه الخوارزمية اكتشفها العالم استرالي اسمة كولن شارنون عام 1986م
- كانت هذه الخوارزمية تستخدم في الإحصاء ولكنها اخذت للتنقيب لأنها تستنبط المعرفة من قواعد بيانات كبيرة.
- الهدف منها هو اننا نبني شجرة قرار و استفيد منها في اتخاذ القرارات الصحيحة.
- أي شجرة معها جذر root اين من هذه الصفات تكون جذر؟ لاحقا تعرف الإجابة

ID3 Algorithm

- قام كولن بتحويل المعلومات الى قيم رقمية بحيث يمكننا التعامل معها في الحاسوب
- عرف المعلومات بانها عبارة عن احتمالية حدوث الشيء

$$entropy(S) = -\sum_{i=1}^k p(i/t) \cdot \log_2 p(i/t)$$

- عرف p انه هي اقل من الواحد لماذا لأنه $\log p(i/t)$ دائما الاحتمالية حقها اقل من واحد وما فيش معنا احتمالية سالبة لذلك نضطر اضربها في سالب
- P تعني فضاء العينة للـ p قسمة فضاء المشكلة
- احنا هنا نتعامل مع اللوغاريتم الثنائي الذي اساسه 2 لانه كولن عرف المعلومات على انها عبارة عن مجموعة من البتات bits

The Process

- Take all unused attributes and calculates their entropies.
- Chooses attribute that has the lowest entropy is minimum or when information gain is maximum
- Makes a node containing that attribute
- خذ جميع الصفات غير المستخدمة وحساب entropies الخاصة بهم.
- يختار الصفة التي تحتوي على أقل entropies هي الحد الأدنى أو عندما يكون كسب المعلومات information gain هو الحد الأقصى
- يجعل العقدة تحتوي على تلك الصفة

ID3 Algorithm

- Algorithm used to generate a decision tree.
 - ID3 is a precursor to the C4.5 Algorithm.
- 1) Establish Classification Attribute (in Table R)
 - 2) Compute Classification Entropy.
 - 3) For each attribute in R, calculate Information Gain using classification attribute.
 - 4) Select Attribute with the highest gain to be the next Node in the tree (starting from the Root node).
 - 5) Remove Node Attribute, creating reduced table R_S .
 - 6) Repeat steps 3-5 until all attributes have been used, or the same classification value remains for all rows in the reduced table.

What is the Entropy ?

- Entropy is an information-theoretic measure of the 'uncertainty' contained in a training set, due to the presence of more than one possible classification.
 - Entropy measures the degree of uniformity or non uniformity in a collection.
- هو مقدار التشوه او العشوائية او عدم النقاء في البيانات.
 - او هو مقياس اقل عدد من bits احتاجها لتشفير هذه المعلومات.

Entropy cont ..



- a training set, due to the presence of more than one possible classification.
- If there are K classes, we can denote the proportion of instances with classification i by p_i for $i = 1$ to K . The value of p_i is the number of occurrences of class i divided by the total number of instances, which is a number between 0 and 1 inclusive.
- The entropy of the training set is denoted by S . It is measured in 'bits' of information.

Computing information

- Information is measured in *bits*
 - Given a probability distribution, the info required to predict an event is the distribution's *entropy*
 - Entropy gives the information required in bits (this can involve fractions of bits!)

- Formula for computing the entropy:

$$entropy(S) = - \sum_{i=1}^k p(i/t) \cdot \log_2 p(i/t)$$

- NOTE: $p(i/t)$ is the relative frequency of class j at (Attribute) node t
- summed over the non-empty classes only, i.e. classes for which

$p_i \neq 0$.

Examples

- Calculate the entropy
- Given:
- Set S contains 14 examples
- 9 Positive values
- 5 Negative values
- $\text{Entropy}(S) = - (9/14) \log_2 (9/14) - (5/14) \log_2 (5/14)$
- $= 0.940$

Information Gain

- هو مقدار الكسب في المعلومات
- هو مقياس لصفة كويسه في عملية التنبؤ لقيم الكلاس
 - يعني الصفة الذي حقها information gain مرتفع اضعه Rule لأنه راح يساعدني في عملية التنبؤ الصحيح
 - كلما كانت information gain قيمته اعلى ما يمكن معناه ان هذا ال Attribute راح يعطيني نتائج افضل في عملية التنبؤ .
 - اذن الصفة الذي حقها information gain اعلى ما يمكن نجعله جذر لأنه من المتوقع انه راح يعطيني تنبؤ اعلا ما يمكن.

Information Gain



- As we mentioned before, to minimize the decision tree depth, when we traverse the tree path, we need to select the optimal attribute for splitting the tree node, which we can easily imply that the attribute with the most entropy reduction is the best choice.
- We define information gain as the expected reduction of entropy related to specified attribute when splitting a decision tree node.
- Information gain is based on the decrease in entropy after a dataset is split on an attribute . Constructing a decision tree is all about finding attribute that returns the highest information gain (i.e., the most homogeneous branches).

Computing the information gain

- Information gain:
(information before split) – (information after split)
- $\text{Gain}(S,X) = \text{Entropy}(S) - \text{Entropy}(S,X)$.
- i.e : attribute X
- Note : Choose attribute with the largest information gain as the decision node.

Example: attribute “Outlook”, 1

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

Compute Classification Entropy(Play)

Play	
Yes	NO
9	5

- $\text{Entropy}(S) = \text{entropy}(9/14, 5/14) = -9/14 \log_2(9/14) - 5/14 \log_2(5/14) = 0.940 \text{ bits}$

Note:

$\log(x) = \ln(x) / \ln 2$.

- $$= (- (9/14 * (\ln((9/14)) / \ln 2)) - (5/14 * (\ln((5/14)) / \ln 2)))$$
$$= 0.940$$

Example: attribute "Outlook"

- "Outlook" = "Sunny": Yes=2, No=3
- $\text{Info}([2,3]) = \text{entropy}(2/5, 3/5) = -2/5 \log_2(2/5) - 3/5 \log_2(3/5) = 0.971$ bits
- "Outlook" = "Overcast": Yes=4, No=0
- $\text{Info}([4,0]) =$
- $\text{entropy}(4/4, 0/5) = -4/4 \log_2(4/4) - 0/4 \log_2(0/4) = 0$ bits
- "Outlook" = "Rainy": Yes=3, No=2
- $\text{Info}([3,2]) = \text{entropy}(3/5, 2/5) = -3/5 \log_2(3/5) - 2/5 \log_2(2/5) = 0.971$ bits
- Expected information for attribute:
- $\text{Info}([3,2], [4,0], [3,2]) =$
 $(5/14) * 0.971 + (4/14) * 0 + (5/14) * 0.971 = 0.693$ bits

*Note: $\log(0)$ is not defined, but we evaluate $0 * \log(0)$ as zero*

Computing the information gain

- Information gain:
(information before split) – (information after split)

$$\begin{aligned}\text{Gain(" Outlook ")} &= \\ \text{info}([9,5]) - \text{Info}([2,3], ([4,0], ([3,2])) &= 0.940 - 0.693 \\ &= 0.247 \text{ bits}\end{aligned}$$

Example: attribute "Temperature", 1

- "Temperature" = "hot": Yes=2, No=2
- $\text{Info}([2,2]) = \text{entropy}(2/4, 2/4) = -2/4 \log_2(2/4) - 2/4 \log_2(2/4) = 1 \text{ bits}$
- "Temperature" = "mild": Yes=4, No=2
- $\text{Info}([4,2]) = \text{entropy}(4/6, 2/6) = -4/6 \log_2(4/6) - 2/6 \log_2(2/6) = 0.918 \text{ bits}$
- "Temperature" = "cool": Yes=3, No=1
- $\text{Info}([3,1]) = \text{entropy}(3/4, 1/4) = -3/4 \log_2(3/4) - 1/4 \log_2(1/4) = 0.811 \text{ bits}$
- Expected information for attribute:
- $\text{Info}([2,2], ([4,2], ([3,1])) = (4/14) * 1 + (6/14) * 0.918 + (4/14) * 0.811$
- $= 0.911 \text{ bits}$

Computing the information gain

- Information gain:
(information before split) – (information after split)
- Gain("Temperature ")=
 $\text{info}([9,5]) - \text{Info}([2,2], ([4,2], ([3,1])) = 0.940 - 0.911$

 $= 0.029 \text{ bits}$

Example: attribute "Humidity"

- "Humidity" = "High": Yes=4, No=3
- $\text{Info}([3,4]) = \text{entropy}(3/7, 4/7) = -3/7 \log_2(3/7) - 4/7 \log_2(4/7) = 0.985 \text{ bits}$
- "Humidity" = "Normal": Yes=6, No=1
- $\text{Info}([6,1]) = \text{entropy}(6/7, 1/7) = -6/7 \log_2(6/7) - 1/7 \log_2(1/7) = 0.592 \text{ bits}$
- Expected information for attribute:
- $\text{Info}([3,4], [6,1]) = (7/14) * 0.985 + (7/14) * 0.592 = 0.79 \text{ bits}$
- Information Gain:

$$\begin{aligned} \text{Gain(" Humidity ")} &= \text{info}([9,5]) - \text{Info}([3,4], ([6,1])) = 0.940 - 0.788 \\ &= 0.152 \text{ bits} \end{aligned}$$

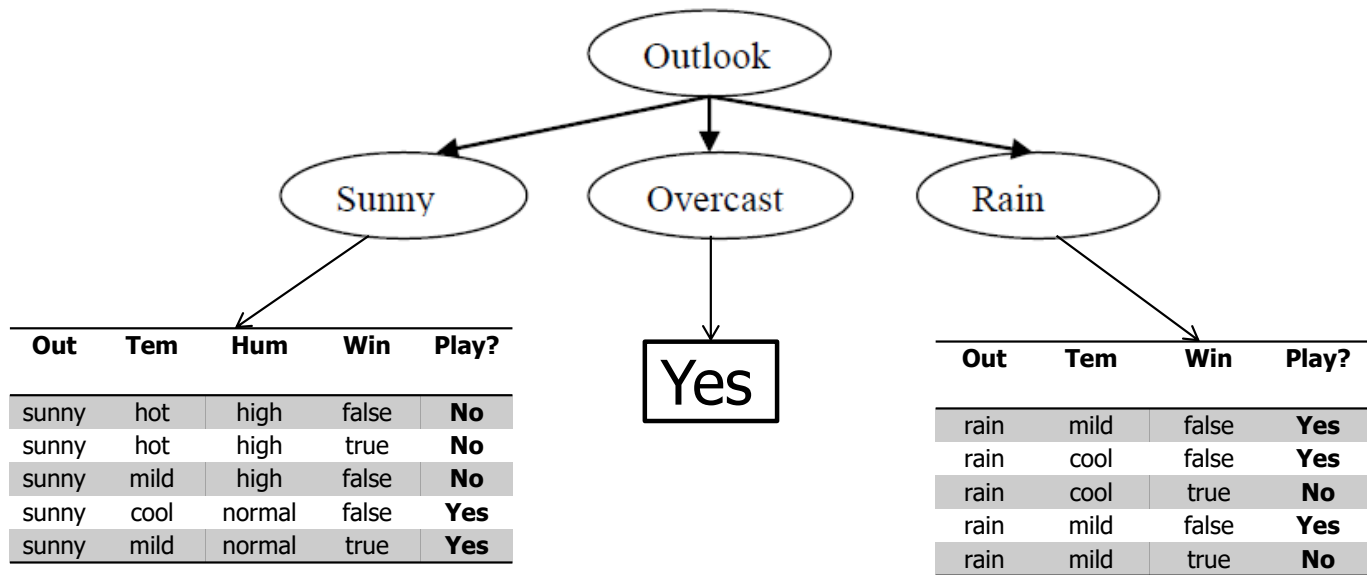
Example: attribute "Windy", 2

- "Windy" = "true": Yes=3, No=3
- $\text{Info}([3,3]) = \text{entropy}(3/6, 3/6) = -3/6 \log_2(3/6) - 3/6 \log_2(3/6) = 1 \text{ bits}$
- "Windy" = "false": Yes=6, No=2
- $\text{Info}([6,2]) = \text{entropy}(6/8, 2/8) = -6/8 \log_2(6/8) - 2/8 \log_2(2/8) = 0.892 \text{ bits}$
- Expected information for attribute:
- $\text{Info}([3,3], ([6,2]) = (6/14) * 1 + (8/14) * 0.811$
- $= 0.892 \text{ bits}$
- Information gain:
- $\text{Gain}(\text{"Windy"}) =$
 $\text{info}([9,5]) - \text{Info}([3,3], ([6,2]) = 0.940 - 0.892$
 $= 0.048 \text{ bits}$

Computing the information gain

- Information gain for attributes from weather data:
- $\text{Gain}(\text{" Outlook"}) = 0.247 \text{ bits}$
- $\text{Gain}(\text{"Temperature"}) = 0.029 \text{ bits}$
- $\text{Gain}(\text{"Humidity"}) = 0.152 \text{ bits}$
- $\text{Gain}(\text{"Windy"}) = 0.048 \text{ bits}$

Example – After Root Node Creation



Example: Choosing the Level 2 Node

- Compute Classification Entropy(**sunny**).

sunny	
Yes	NO
2	3

- $\text{entropy}(\mathbf{sunny}) = \text{entropy}(2/5, 3/5) =$
- $-2/5 \log_2(2/5) - 3/5 \log_2(3/5) = \mathbf{0.9711 \text{ bits}}$

Example: attribute “Temperature” with sunny , 1

- Attribute temperature Temp value can be hot, mild or cool.
- **sunny**(hot)= hot is of occurrences 2
 - yes=0, no=2
- **sunny**(mild)= mild is of occurrences 2
 - yes=1, no=1
- **sunny**(cool)= cool is of occurrences 1
 - yes=1, no=0

Example: attribute “Temperature” with sunny , 2

- Entropy (**sunny**, Temp (hot))=0
- Entropy (**sunny**, Temp (mild))=1
- Expected information for attribute Temp (mild):
- $\text{Info}(\mathbf{sunny}, \text{Temp (mild)}) = (2/5) * 1 = 0.4$
- Entropy (sunny, Temp (cool))=0
- Gain (**sunny**, Temp) = $0.971 - 0 + 0.4 + 0 = 0.571$ bits

Example: attribute “Humidity” with sunny, 1

- Attribute Humidity Hum value can be high, normal.
- **sunny**(Hum (high))= high is of occurrences 3
 - yes=0, no=3
- **sunny**(Hum (normal))= normal is of occurrences 2
 - yes=2, no=0
- Entropy (**sunny**, Hum (high))=0
- Entropy (**sunny**, Hum (normal))=0
- Gain (**sunny**, Hum) =0.971-0+0=**0.971 bits**

Example: attribute “Windy” with sunny,1

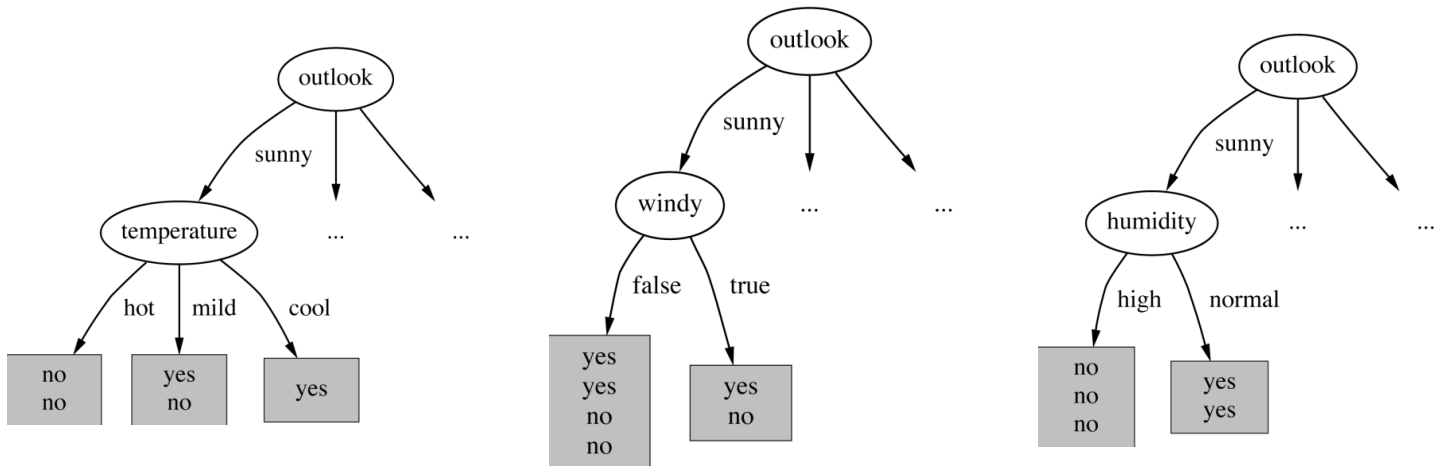
- Attribute Windy Wind value can be true, false.
- **sunny**(Wind (true))= true is of occurrences 2
 - yes=1, no=1
- **sunny**(Wind (false))= false is of occurrences 3
 - yes=1, no=2
- Entropy (**sunny**, Wind (true))=1
- Expected information for attribute Wind (true):
- Info(**sunny**, Wind (true))=(2/5)*1=0.4

Example: attribute “Windy” with sunny,2

- Entropy (**sunny**, Wind (false))=
- $-1/3\log_2(1/3)-2/3\log_2(2/3) = 0.918$ bits
- Expected information for attribute Wind (true):
- $\text{Info}(\mathbf{sunny}, \text{Wind (false)}) = (3/5) * 0.918 = 0.550$ bits

- $\text{Gain}(\mathbf{sunny}, \text{Wind}) = 0.971 - 0.4 + 0.550 = \mathbf{0.021 \text{ bits}}$

Continuing to split



$\text{gain}(\text{"Humidity"}) = 0.971 \text{ bits}$

$\text{gain}(\text{"Temperature"}) = 0.571 \text{ bits}$

$\text{gain}(\text{"Windy"}) = 0.020 \text{ bits}$

Example: Choosing the Level 2 Node

- Compute Classification Entropy(**rain**).

rain	
Yes	NO
3	2

- $\text{entropy}(\mathbf{rain}) = \text{entropy}(3/5, 2/5) =$
- $-3/5 \log_2(3/5) - 2/5 \log_2(2/5) = 0.9711 \text{ bits}$

Example: attribute "Temperature" with rain , 1

- Attribute temperature Temp value can be mild or cool.
- **rain** (mild)= mild is of occurrences 3
 - yes=2, no=1
- **rain** (cool)= cool is of occurrences 2
 - yes=1, no=1
- Entropy (**rain** , Temp (mild))=
- $-2/3\log_2(2/3)-1/3\log_2(1/3) = 0.918$ bits
- Expected information for attribute Temp (mild):
- $\text{Info}(\mathbf{rain} , \text{Temp (mild)}) = (3/5) * 0.918 = 0.550$ bits

Example: attribute “Temperature” with rain , 2

- Entropy (**rain** , Temp (cool))=1
- Expected information for attribute Temp (cool):
- Info(**rain** , Temp (cool))=(2/5)*1=0.4
- Gain (**rain**, Temp) =0.971-0.4+0.550=**0.021 bits**

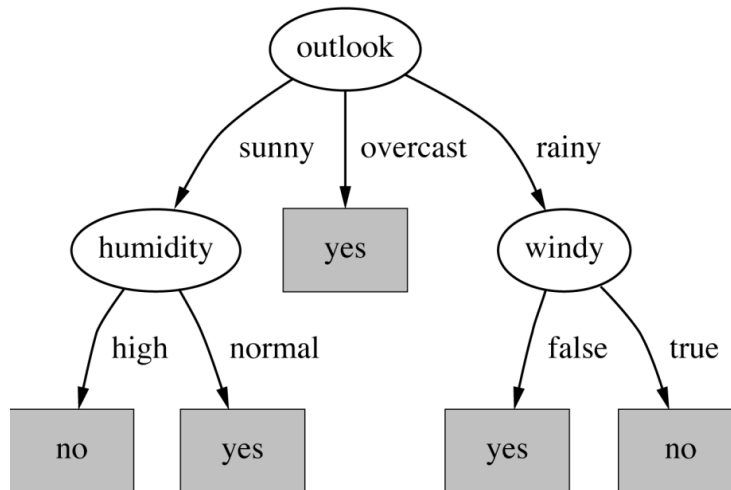
Example: attribute “Windy” with rain, 1

- Attribute Windy Wind value can be true, false.
- **rain** (true)= true is of occurrences 2
 - yes=0, no=2
- **rain** (false)= false is of occurrences 3
 - yes=3, no=0
- Entropy (**rain** , Wind (true))=0
- Expected information for attribute Wind (true):
- Info(**rain** , Wind (true))=(2/5)*0=0 bits

Example: attribute “Windy” with rain, 2

- Entropy (**rain** , Wind (false))=1
- Expected information for attribute Wind (true):
- Info(**rain** , Temp (mild))=(3/5)*0=0 bits
- Gain (**rain**, Wind) =0.971-0 +0=**0.971 bits**

The final decision tree



- Note: not all leaves need to be pure; sometimes identical instances have different classes
⇒ Splitting stops when data can't be split any further

Data Mining



gain ratio and Prism Algorithm

Generating decision rules

The decision tree can also be expressed in rule format:

- IF outlook = sunny AND humidity = high THEN play = no
- IF outlook = sunny AND humidity = normal THEN play = yes
- IF outlook = overcast THEN play = yes
- IF outlook = rain AND wind = true THEN play = no
- IF outlook = rain AND wind = false THEN play = yes

صفات كثيرة التفرع

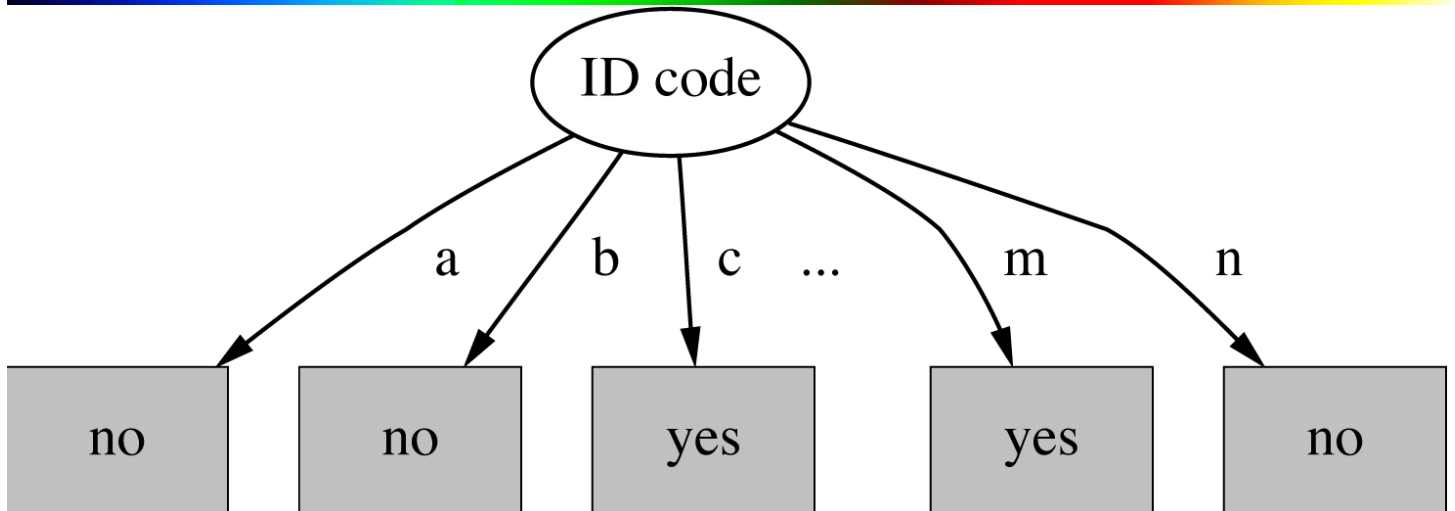
Highly-branching attributes

- المشكلة: الصفات معها عدد كبير من القيم (مثل حالة: ID code)
- Problematic: attributes with a large number of values (extreme case: ID code)
 - من المرجح أن تكون المجموعات الفرعية نقية إذا كان هناك عدد كبير من القيم
- Subsets are more likely to be pure if there is a large number of values
 - ⇒ ينحاز Information gain نحو اختيار السمات التي معها عدد كبير من القيم
 - ⇒ Information gain is biased towards choosing attributes with a large number of values
 - ⇒ قد يؤدي ذلك إلى *overfitting* (اختيار سمة غير مثالية للتنبؤ) يعني تمثيل الـ DataSet نفسها.
 - ⇒ This may result in *overfitting* (selection of an attribute that is non-optimal for prediction)

Weather Data with ID code

ID	Outlook	Temperature	Humidity	Windy	Play?
A	sunny	hot	high	false	No
B	sunny	hot	high	true	No
C	overcast	hot	high	false	Yes
D	rain	mild	high	false	Yes
E	rain	cool	normal	false	Yes
F	rain	cool	normal	true	No
G	overcast	cool	normal	true	Yes
H	sunny	mild	high	false	No
I	sunny	cool	normal	false	Yes
J	rain	mild	normal	false	Yes
K	sunny	mild	normal	true	Yes
L	overcast	mild	high	true	Yes
M	overcast	hot	normal	false	Yes
N	rain	mild	high	true	No

Split for ID Code Attribute



Entropy of split = 0 (since each leaf node is "pure", having only one case).

Information gain is maximal for ID code

Gain ratio

- *Gain ratio* : تعديل information gain التي تقلل من انحيازها على سمات كثيرة التفرع
- *Gain ratio*: a modification of the information gain that reduces its bias on high-branch attributes
- Gain ratio يجب أن تكون
- Gain ratio should be
 - كبير عندما يتم توزيع البيانات بالتساوي
 - Large when data is evenly spread
 - صغير عندما تنتمي جميع البيانات إلى فرع واحد
 - Small when all data belong to one branch

Gain ratio

- يأخذ Gain ratio عدد وحجم الفروع في الاعتبار عند اختيار سمة
- Gain ratio takes number and size of branches into account when choosing an attribute
- بتصحيح information gain عن طريق أخذ المعلومات الجوهرية *intrinsic information* للانقسام split في الاعتبار (أي كم المعلومات التي نحتاجها لمعرفة أي فرع ينتمي إليه instance)
- It corrects the information gain by taking the *intrinsic information* of a split into account (i.e. how much info do we need to tell which branch an instance belongs to)

Gain Ratio and Intrinsic Info.

■ Intrinsic information: إنتروبي توزيع الحالات (السجلات) إلى فروع

- Intrinsic information: entropy of distribution of instances into branches

$$\text{IntrinsicInfo}(S, A) \equiv -\sum \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|}.$$

- *Gain ratio* (Quinlan'86) normalizes info gain by:

$$\text{GainRatio}(S, A) = \frac{\text{Gain}(S, A)}{\text{IntrinsicInfo}(S, A)}.$$

Computing the gain ratio

- Example: intrinsic information for ID code

$$\text{info}([1,1,\dots,1) = 14 \times (-1/14 \times \log 1/14) = 3.807 \text{ bits}$$

- **Importance of attribute decreases as intrinsic information gets larger**

- Example of gain ratio:

$$\text{gain_ratio}(\text{"Attribute"}) = \frac{\text{gain}(\text{"Attribute"})}{\text{intrinsic_info}(\text{"Attribute"})}$$

- Example:

$$\text{gain_ratio}(\text{"ID_code"}) = \frac{0.940 \text{ bits}}{3.807 \text{ bits}} = 0.246$$

Example: Computing the gain ratio(Outlook)

1- compute information for attribute

- "Outlook" = "Sunny": Yes=2, No=3

$$\text{Info}([2,3]) = \text{entropy}(2/5, 3/5) = -2/5 \log_2(2/5) - 3/5 \log_2(3/5) = 0.971 \text{ bits}$$

- "Outlook" = "Overcast": Yes=4, No=0

$$\text{Info}([4,0]) =$$

$$\text{entropy}(4/4, 0/4) = -4/4 \log_2(4/4) - 0/4 \log_2(0/4) = 0 \text{ bits}$$

- "Outlook" = "Rainy": Yes=3, No=2

$$\text{Info}([3,2]) = \text{entropy}(3/5, 2/5) = -3/5 \log_2(3/5) - 2/5 \log_2(2/5) = 0.971 \text{ bits}$$

- Expected information for attribute:

$$\text{Info}([3,2], [4,0], [2,3]) =$$

$$(5/14) * 0.971 + (4/14) * 0 + (5/14) * 0.971 = 0.693 \text{ bits}$$

Computing the information gain

2- compute Information gain:

Gain(" Outlook ")=

$$\text{info}([9,5]) - \text{Info}([2,3], ([4,0],[3,2]) = 0.940 - 0.693$$

$$= 0.247 \text{ bits}$$

Computing the gain ratio(Outlook)

3- compute intrinsic information for Outlook

$\text{info}([5,4,5]) =$

$$-5/14 \log_2(5/14) - 4/14 \log_2(4/14) - 5/14 \log_2(5/14) = 1.577 \text{ bits}$$

- $(-(5/14 * (\ln((5/14))/\ln 2)) - (4/14 * (\ln((4/14))/\ln 2)) - (5/14 * (\ln((5/14))/\ln 2))) = 1.577$
bits

- Example of gain ratio:

$$\text{gain_ratio}(\text{"Attribute"}) = \frac{\text{gain}(\text{"Attribute"})}{\text{intrinsic_info}(\text{"Attribute"})}$$

- Example:

$$\text{gain_ratio}(\text{"outlook"}) = \frac{0.247 \text{ bits}}{1.1577 \text{ bits}} = 0.156$$

Gain ratios for weather data

Outlook		Temperature	
Info:	0.693	Info:	0.911
Gain: $0.940 - 0.693$	0.247	Gain: $0.940 - 0.911$	0.029
Split info: $\text{info}([5,4,5])$	1.577	Split info: $\text{info}([4,6,4])$	1.362
Gain ratio: $0.247/1.577$	0.156	Gain ratio: $0.029/1.362$	0.021

Humidity		Windy	
Info:	0.788	Info:	0.892
Gain: $0.940 - 0.788$	0.152	Gain: $0.940 - 0.892$	0.048
Split info: $\text{info}([7,7])$	1.000	Split info: $\text{info}([8,6])$	0.985
Gain ratio: $0.152/1$	0.152	Gain ratio: $0.048/0.985$	0.049

خوارزمية النشر Prism Algorithm

- هي إحدى خوارزميات classification
- تعتمد على الاحتمالات probability حيث تأخذ أعلى احتمال
- تقوم الخوارزمية بعملية النشر بأخذ كل قيم الصفات و تصنفها على إحدى قيم الكلاس ، و تستخرج القواعد حتى تصل الي تصنيف واحد و بعدها تكرر النشر على قيمة الكلاس الثاني و هكذا
- نقوم بعملية النشر وفق الجدول التالي

قيم الصفات	عدد تكرار القيمة لهذا الكلاس	عدد تكرار هذه القيمة على مستوى قاعدة البيانات	حساب الاحتمالات كل قيمة

- تحسب الاحتمالية وفق القانون التالي
عدد تكرار القيمة لهذا الكلاس / قسمة عدد تكرار هذه القيمة على مستوى قاعدة البيانات

Prism Algorithm

Outlook	Temperature	Humidity	Wind	PlayTennis?	
sunny	hot	high	weak	no	
sunny	hot	high	strong	no	
overcast	hot	high	weak	yes	
rainy	mild	high	weak	yes	
rainy	cool	normal	weak	yes	
rainy	cool	normal	strong	no	
overcast	cool	normal	strong	yes	
sunny	mild	high	weak	no	
sunny	cool	normal	weak	yes	
rainy	mild	normal	weak	yes	
sunny	mild	normal	strong	yes	
overcast	mild	high	strong	yes	
overcast	hot	normal	weak	yes	
rainy	mild	high	strong	no	

Class Play=Yes, step 1

Attribute/Value Pair	Frequency for Class=Yes	Total Freq. out of 14 Instances	Probability
Sunny	2	5	0.4
Overcast	4	4	1
Rainy	3	5	0.6
Hot	2	4	0.5
Mild	4	6	0.67
Cool	3	4	0.75
High	3	7	0.43
Normal	6	7	0.86
Weak	6	8	0.75
Strong	3	6	0.5

- Max. Prob.=1
- Incomplete Rule1: **If** Outlook=Overcast **Then** Play=Yes.

Class Play=Yes, step 1

- نشر قاعدة البيانات على الكلاس Yes
- حساب احتمالية كل قيمة مثلا Sunny عدد وجودها في الكلاس yes هو 2 و عدد وجودها في قاعدة البيانات هو 5 نحسب الاحتمالية
- $2/5=0.4$
- نكرر العملية على باقي القيم
- نختار اعلى احتمالية مثلا طلع في مثالنا العدد 1 للقيمة Overcast
- نكون قاعدة غير مكتملة (أولية) لهذه القيمة
- If Outlook=Overcast Then Play=Yes.
- نقوم باستخراج جميع السجلات التي تحتويها القاعدة 1 كما يلي:

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 1 كما يلي:

- Extract all instances which covered by Incomplete Rule1 as follows:

Outlook	Temp	Hum	Wind	Play
Overcast	Hot	High	Weak	Yes
Overcast	Cool	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes

- تحتوي هذه المجموعة الفرعية فقط على سجلات الكلاس Play=Yes
- This Subset contains only instances of class Play=Yes.
- ولذلك فإن القاعدة النهائية (القاعدة 1) هي:
- The final induced rule(Rule 1) is therefore:
- **If** Outlook=Overcast **Then** Play=Yes

Removing the four instances covered by the first rule (Rule 1) from the training set gives a new training set with 10 instances as follows:

■ وإزالة الاربعة السجلات التي تحتويها القاعدة الأولى (القاعدة 1) من مجموعة التدريب training set يعطي مجموعة تدريب جديدة مع 10 سجلات على النحو التالي:

Outlook	Temp	Hum	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Rainy	Mild	High	Strong	No

Attribute/Value Pair	Frequency for Class=Yes	Total Freq. out of 10 Instances	Probability
Sunny	2	5	0.4
Rainy	3	5	0.6
Hot	0	2	0
Cool	2	3	0.67
Mild	3	5	0.6
High	1	4	0.25
Normal	4	5	0.80
Weak	4	6	0.67
Strong	1	4	0.25

- Max. Prob.=0.80
- Incomplete Rule2: **If** Humidity=Normal **Then** Play=Yes.

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 2 كما يلي:

- Extract all instances which covered by Incomplete Rule2 as follows:

Outlook	Temp	Humi	Wind	Play
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Attribute	Freq. Play=Yes	Total Freq.	Probability
Rainy	2	3	0.67
Sunny	2	2	1
Cool	2	3	0.67
Mild	2	2	1
Normal	4	5	0.8
Weak	3	3	1
Strong	1	2	0.50

نقوم بنشر السجلات المستخرجة للكلاس yes واختيار اعلى احتمالية و كتابة القاعدة 2 بشكل التالي.

- Max. Prob.=1
- The Incomplete Rule2 is :
- **If** Humidity=Normal AND Outlook=Sunny **Then** Play=Yes

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 2 من السجلات التي تم استخراجها في الخطوة السابقة كما يلي:

- Extract all instances which covered by Incomplete Rule2 as follows:

Outlook	Temp	Hum	Wind	Play
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

- This Subset contains only instances of class Play=Yes.
- The final induced rule(Rule2) is therefore:
- **If** Humidity=Normal AND Outlook=Sunny **Then** Play=Yes

وإزالة السجلين الذي تحتويها القاعدة 2 تعطي مجموعة تدريب جديدة تضم 8 سجلات على النحو التالي:

- Removing the two instances covered by the Rule2 gives a new training set with 8 instances as follows:

Outlook	Temp	Hum	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Sunny	Mild	High	Weak	No
Rainy	Mild	Normal	Weak	Yes
Rainy	Mild	High	Strong	No

نقوم بنشر قاعدة البيانات الجديدة المكونة من 8 سجلات
Class Play=Yes, step 3

Attribute/Value Pair	Frequency for Class=Yes	Total Freq. out of 8 Instances	Probability
Sunny	0	3	0
Rainy	3	5	0.6
Hot	0	2	0
Cool	1	2	0.50
Mild	2	4	0.50
High	1	5	0.20
Normal	2	3	0.67
Weak	3	5	0.60
Strong	0	3	0

- Max. Prob.=0.67
- Incomplete Rule3: If Humi.=Normal Then Play=Yes.

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 3 كما يلي:

Extract all instances which covered by Incomplete Rule3 as follows

Out	Temp	Hum	Wind	Play	Attribute	Freq. Play=Yes	Total Freq.	Probability
Rainy	Cool	Normal	Weak	Yes	Rainy	2	3	0.67
Rainy	Cool	Normal	Strong	No	Mild	1	1	1.00
Rainy	Mild	Normal	Weak	Yes	Cool	1	2	0.50
					Weak	2	2	1
					Strong	0	1	0

نقوم بنشر السجلات المستخرجة للكلاس yes واختيار اعلى احتمالية و كتابة القاعدة 3 بشكل التالي.

- Max. Prob.=1
- Complete Rule3: If Humidity=Normal And Temp.=Mild Then Play=Yes.

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 3 من السجلات التي تم استخراجها في الخطوة السابقة كما يلي:

- Extract all instances which covered by Incomplete Rule3 as follows:

Rainy	Mild	Normal	Weak	Yes
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- This Subset contains only instances of class Play=Yes.
- The final induced rule(Rule3) is therefore:
- **If** Humidity=Normal AND Temp.=Mild **Then** Play=Yes

إزالة السجل الذي تحتويها القاعدة 3 تعطي مجموعة تدريب جديدة تضم 7 سجلات على النحو التالي:

- Removing the one instance covered by the Rule3 gives a new training set with 7 instances as follows:

Outlook	Temp	Hum	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Sunny	Mild	High	Weak	No
Rainy	Mild	High	Strong	No

Class Play=Yes, step 4

Attribute/Value Pair	Frequency for Class=Yes	Total Freq. out of 8 Instances	Probability
Sunny	0	3	0
Rainy	2	4	0.5
Hot	0	2	0
Cool	1	2	0.50
Mild	1	3	0.33
High	1	5	0.20
Normal	1	2	0.50
Weak	2	3	0.67
Strong	0	3	0

- Max. Prob.=0.67
- Incomplete Rule4: If Wind=Weak Then Play=Yes.

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 4 من السجلات التي تم استخراجها في الخطوة السابقة كما يلي:

- Extract all instances which covered by Incomplete Rule4 as follows:

Outlook	Temp	Hum	Wind	Play
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes

- This Subset contains only two instances of class Play=Yes.
- The final induced rule(Rule 4) is therefore:
- **If** Wind=Weak **Then** Play=Yes

إزالة السجلين الذي تحتويها القاعدة 4 تعطي مجموعة تدريب جديدة تضم 5 سجلات على النحو التالي:

Removing the TWO instances covered by the Rule4 gives a new training set with the following instances as follows:

Outlook	Temp	Hum	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Rainy	Cool	Normal	Strong	No
Sunny	Mild	High	Weak	No
Rainy	Mild	High	Strong	No

■ في هذه الخطوة، تمت الآن إزالة جميع سجلات الكلاس class Play=Yes لذلك ينتهي خوارزمية النشر مع القواعد التالية class Play=Yes.

- At this step all instances of class Play=Yes have now been removed. So the Prism algorithm terminates with the following rules for class Play=Yes.
- If Outlook=Overcast Then Play =Yes
- If Humidity=Normal AND Outlook=Sunny Then Play=Yes
- If Humidity=Normal AND Temp.=Mild Then Play=Yes.
- If Wind=Weak Then Play =Yes
- Do the same for the class Play=No.

Class Play=No , step 1

Attribute/Value Pair	Frequency for Class=No	Total Freq. out of 14 Instances	Probability
Sunny	3	5	0.6
Overcast	0	4	0
Rainy	2	5	0.4
Hot	2	4	0.5
Mild	2	6	0.33
Cool	1	4	0.25
High	4	7	0.57
Normal	1	7	0.14
Weak	2	8	0.25
Strong	3	6	0.5

- Max. Prob.=0.6
- Incomplete Rule1: **If** Outlook=Sunny **Then** Play=NO.

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 1 كما يلي:

- Extract all instances which covered by Incomplete Rule1 as follows:

Outlook	Temp	Humi	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Attribute	Freq. Play=No	Total Freq.	Probability
hot	2	2	1
mild	1	2	1
cool	0	1	0.67
High	3	3	1
Normal	0	2	0
Weak	2	3	0.67
Strong	1	2	0.5

نقوم بنشر السجلات المستخرجة للكلاس NO واختيار اعلى احتمالية و كتابة القاعدة 1 بشكل التالي.

- Max. Prob.=1
- The Incomplete Rule1 is :
- **If** Outlook=Sunny AND Temperature= Hot **Then** Play=No

نقوم باستخراج جميع السجلات التي تحتويها القاعدة غير المكتملة 1 كما يلي:

- Extract all instances which covered by Incomplete Rule1 as follows:

Outlook	Temp	Hum	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No

- تحتوي هذه المجموعة الفرعية فقط على سجلات الكلاس Play=No
- This Subset contains only instances of class Play=No.
- ولذلك فإن القاعدة النهائية (القاعدة 1) هي:
- The final induced rule(Rule1) is therefore:
- **If** Outlook=Sunny AND Temperature= Hot **Then** Play=No

Removing the TWO instances covered by the first rule (Rule 1) from the training set gives a new training set with 12 instances as follows:

Outlook	Temp	Hum	Wind	Play
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rainy	Mild	High	Strong	No

Class Play=No , step 2

Attribute/Value Pair	Frequency for Class=No	Total Freq. out of 12 Instances	Probability
Sunny	1	3	0.3
Overcast	0	4	0
Rainy	2	5	0.4
Hot	0	2	0
Mild	2	6	0.33
Cool	1	4	0.25
High	2	5	0.4
Normal	1	7	0.14
Weak	1	7	0.14
Strong	2	5	0.4

- Max. Prob.=0.4
- Incomplete Rule2: **If** Outlook=Rainy **Then** Play=NO.

- Extract all instances which covered by Incomplete Rule2 as follows:

Outlook	Temp	Humi	Wind	Play
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Rainy	Mild	Normal	Weak	Yes
Rainy	Mild	High	Strong	No

Attribute	Freq. Play=No	Total Freq.	Probability
Mild	1	3	0.33
Cool	1	2	0.5
High	1	2	0.5
Normal	1	3	0.33
Weak	0	3	0
Strong	2	2	1

- Max. Prob.=1
- The Incomplete Rule2 is :
- **If** Outlook= Rainy AND Wind = Strong **Then** Play=No

- Extract all instances which covered by Incomplete Rule2 as follows:

Outlook	Temp	Hum	Wind	Play
Rainy	Cool	Normal	Strong	No
Rainy	Mild	High	Strong	No

- This Subset contains only instances of class Play=No.
- The final induced rule(Rule2) is therefore:
- **If** Outlook= Rainy AND Wind = Strong **Then** Play=No

Removing the TWO instances covered by the first rule (Rule2)from the training set gives a new training set with 10instances as follows:



Outlook	Temp	Hum	Wind	Play
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes

Class Play=No , step 3

Attribute/Value Pair	Frequency for Class=No	Total Freq. out of 10 Instances	Probability
Sunny	1	3	0.33
Overcast	0	4	0
Rainy	0	3	0
Hot	0	2	0
Mild	1	5	0.2
Cool	0	4	0
High	1	4	0.25
Normal	0	6	0
Weak	1	7	0.14
Strong	0	2	0

- Max. Prob.=0.33
- Incomplete Rule3: **If** Outlook=Sunny **Then** Play=NO.

- Extract all instances which covered by Incomplete Rule3 as follows:

Outlook	Temp	Humi	Wind	Play
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Attribute	Freq. Play=No	Total Freq.	Probability
Mild	1	2	0.5
Cool	0	1	0
High	1	1	1
Normal	0	2	0
Weak	1	2	0.5
Strong	0	1	0

- Max. Prob.=1
- The Incomplete Rule3 is :
- **If** Outlook=Sunny AND Humidity=High **Then** Play=No.

- 
- Extract all instances which covered by Incomplete Rule3 as follows:

Sunny	Mild	High	Weak	No
--------------	-------------	-------------	-------------	-----------

- This Subset contains only instances of class Play=No.
- The final induced rule(Rule3) is therefore:
- **If** Outlook=Sunny **AND** Humidity=High **Then** Play=No

Removing the One instances covered by the Rule3 gives a new training set with the following instances as follows:



Outlook	Temp	Hum	Wind	Play
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Overcast	Cool	Normal	Strong	Yes
Sunny	Cool	Normal	Weak	Yes
Rainy	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes

- 
- At this step all instances of class Play=No have now been removed. So the Prism algorithm terminates with the following rules for class Play=No.
 - **If** Outlook=Sunny AND Temperature= Hot **Then** Play=No.
 - **If** Outlook=Rainy AND Wind=Strong **Then** Play=No.
 - **If** Outlook=Sunny AND Humidity=High **Then** Play=No.

Data Mining



C4.5 algorithm

C4.5 algorithm

- As C4.5 is an improvement of ID3, then the first step of calculating the gain is the same except for the attributes to continuous values

- Computing the entropy

$$entropy(S) = - \sum_{i=1}^k p_i \cdot \log_2 p_i$$

- Computing the information gain

- Information gain:

(information before split) – (information after split)

Gain(S,X)= Entropy(S)- Entropy(S,X).

Example of C4.5 algorithm

Database T:

TABLE A simple flat database of examples for training

Attribute1	Attribute2	Attribute3	Class
A	70	True	CLASS1
A	90	True	CLASS2
A	85	False	CLASS2
A	95	False	CLASS2
A	70	False	CLASS1
B	90	True	CLASS1
B	78	False	CLASS1
B	65	True	CLASS1
B	75	False	CLASS1
C	80	True	CLASS2
C	70	True	CLASS2
C	80	False	CLASS1
C	80	False	CLASS1
C	96	False	CLASS1

Compute Classification Entropy(Play)

CLASS	
CLASS1	CLASS2
9	5

- $\text{Entropy}(S) = \text{entropy}(9/14, 5/14) = -9/14 \log_2(9/14) - 5/14 \log_2(5/14) = 0.940 \text{ bits}$

Note:

$\log(x) = \ln(x) / \ln(2)$.

- $= (- (9/14 * (\ln((9/14)) / \ln(2))) - (5/14 * (\ln((5/14)) / \ln(2)))) = 0.940 \text{ bits}$

Example: attribute "Attribite1"

- "Attribite1" = "A": CLASS1 =2, CLASS2 =3
 - $\text{Info}([2,3]) = \text{entropy}(2/5, 3/5) = -2/5 \log_2(2/5) - 3/5 \log_2(3/5) = 0.971 \text{ bits}$
 - "Attribite1" = "B": CLASS1 =4, CLASS2=0
 - $\text{Info}([4,0]) =$
 - $\text{entropy}(4/4, 0/5) = -4/4 \log_2(4/4) - 0/4 \log_2(0/4) = 0 \text{ bits}$
 - "Attribite1" = "C": CLASS1 =3, CLASS2=2
 - $\text{Info}([3,2]) = \text{entropy}(3/5, 2/5) = -3/5 \log_2(3/5) - 2/5 \log_2(2/5) = 0.971 \text{ bits}$
 - Expected information for attribute:
 - $\text{Info}([3,2], [4,0], [3,2]) =$
 $(5/14) * 0.971 + (4/14) * 0 + (5/14) * 0.971 = 0.693 \text{ bits}$
 - Information gain:
- $\text{Gain}(\text{"Attribite1"}) =$
 $\text{info}([9,5]) - \text{Info}([2,3], ([4,0], [3,2])) = 0.940 - 0.693 = 0.247 \text{ bits}$

Example: attribute "Attribite3"

- "Attribite3" = "true": CLASS1 =3, CLASS2=3
- $\text{Info}([3,3]) = \text{entropy}(3/6, 3/6) = -3/6 \log_2(3/6) - 3/6 \log_2(3/6) = 1 \text{ bits}$
- "Attribite3" = "false": CLASS1 =6, CLASS2=2
- $\text{Info}([6,2]) = \text{entropy}(6/8, 2/8) = -6/8 \log_2(6/8) - 2/8 \log_2(2/8) = 0.892 \text{ bits}$
- Expected information for attribute Attribite3 :
- $\text{Info}([3,3], ([6,2]) = (6/14) * 1 + (8/14) * 0.811 = 0.892 \text{ bits}$
- Information gain:
- $\text{Gain}(\text{" Attribite3 "}) = \text{info}([9,5]) - \text{Info}([3,3], ([6,2]) = 0.940 - 0.892 = 0.048 \text{ bits}$

C4.5 algorithm

- C4.5 contains mechanisms for proposing three types of tests:
 - The “standard” test on a discrete attribute, with one outcome and branch for each possible value of that attribute.
 - If attribute Y has **continuous numeric values**, a binary test with outcomes $Y \leq Z$ and $Y > Z$ could be defined, based on comparing the value of attribute against a **threshold value Z**.
 - A more complex test based also on a discrete attribute, in which the possible values are allocated to a variable number of groups with one outcome and branch for each group.

Handle numeric values



- Threshold value Z :
 - The training samples are first sorted on the values of the attribute Y being considered. There are only a **finite number of these values**, so let us denote them in sorted order as $\{v_1, v_2, \dots, v_m\}$.
 - Any threshold value lying between v_i and v_{i+1} will have the same effect of dividing the cases into those whose value of the attribute Y lies in $\{v_1, v_2, \dots, v_i\}$ and those whose value is in $\{v_{i+1}, v_{i+2}, \dots, v_m\}$. There are thus only $m-1$ possible splits on Y , all of which should be examined systematically to obtain an optimal split.

Handle numeric values



- It is usual to choose the midpoint of each interval: $(v_i + v_{i+1})/2$ as the representative threshold.
- C4.5 chooses as the threshold a smaller value v_i for every interval $\{v_i, v_{i+1}\}$, rather than the midpoint itself.

Example : calculation of information gain for an attribute of continuing value.

- calculation of information gain for an attribute (Attribute2):

1- After a sorting process, the set of values is:

$\{65, 70, 70, 70, 75, 78, 80, 80, 80, 85, 90, 90, 95, 96\}$

2- we will remove values that are repeated:

$\{65, 70, 75, 78, 80, 85, 90, 95, 96\}$.

3- the set of potential threshold values Z is (C4.5):

$\{65, 70, 75, 78, 80, 85, 90, 95, 96\}$.

GAIN CALCULATION FOR THE ATTRIBUTE CONTINUOUS Attribute2 USING C4.5 ALGORITHM

	65		70		75		78		80		85		90		95		96	
interval	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
CLASS1	1	8	3	6	4	5	5	4	7	2	7	2	8	1	8	1	9	0
CLASS2	0	5	1	4	1	4	1	4	2	3	3	2	4	1	5	0	5	0
Entropy	0	0.961	0.811	0.971	0.721	0.911	0.65	1	0.764	0.971	0.881	1	0.918	1	0.961	0	0.94	0
Info(S, T)	0.892		0.925		0.8950		0.85		0.838		0.915		0.929		0.892		0.94	
Gain	0.048		0.015		0.045		0.09		0.102		0.025		0.011		0.045		0	

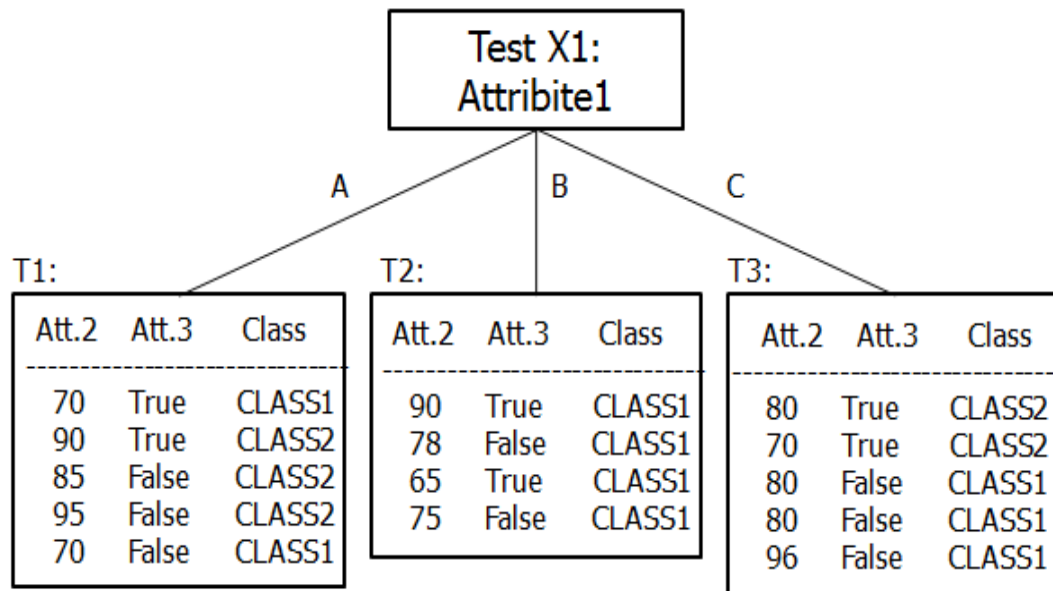
GAIN CALCULATION FOR THE ATTRIBUTE CONTINUOUS Attribute2 USING C4.5 ALGORITHM

- Entropy(≤ 65) = 0
- Entropy(> 65) = $-8/13 \log_2(8/13) - 5/13 \log_2(5/13) = 0.961$ bits
 $(-(8/13 * (\ln((8/13)) / \ln 2)) - (5/13 * (\ln((5/13)) / \ln 2))) = 0.961$ bits
- Info(S,T) = $(1/14) * 0 + (13/14) * 0.961 = 0.892$ bits
- Gain = $0.940 - 0.892 = 0.048$ bits

Example: Choosing the Level 2 Node

- $\text{Gain}(\text{Attribute1}) = 0.940 - 0.694 = 0.246$ bits
- $\text{Gain}(\text{Attribute2}) = 0.940 - 0.837 = 0.103$ bits
- $\text{Gain}(\text{Attribute3}) = 0.940 - 0.892 = 0.048$ bits
- Attribute1 gives the highest gain of 0.246 bits, and therefore this attribute will be selected for the first splitting.

Example of C4.5 algorithm



Compute Classification Entropy(Attribute1)

- Compute Classification Entropy(Attribute1).

Attribute1	
CLASS1	CLASS2
2	3

- $\text{entropy}(\text{Attribute1}) = \text{entropy}(2/5, 3/5) =$
- $-2/5 \log_2(2/5) - 3/5 \log_2(3/5) = 0.9711 \text{ bits}$

- Compute Classification Entropy(Attribute3).

Attribute3	
true	false
2	3

- $\text{entropy}(\text{Attribute1}, \text{Attribute3}(\text{true})) = \text{entropy}(1/2, 1/2) = 1$
- Entropy (Attribute1, Attribute3 (false))=

 $-1/3\log_2(1/3) - 2/3\log_2(2/3) = 0.918 \text{ bits}$
- Expected information for attribute Attribute3 :
- $\text{Info}(\text{Attribute1}, \text{Attribute3}) = (2/5)*1 + (3/5)*0.918 = 0.4 + 0.550 = 0.950 \text{ bits}$
- $\text{Gain}(\text{Attribute1}, \text{Attribute3}) = 0.971 - 0.950 = 0.021 \text{ bits}$

Example : calculation of information gain for an attribute of continuing value.

- Attribute2:
- After a sorting process, the set of values is:
 $\{70, 70, 85, 90, 95\}$
we will remove values that are repeated:
 $\{\{70, 70, 85, 90, 95\}\}$
the set of potential threshold values Z is (C4.5):
 $\{\{70, 70, 85, 90, 95\}\}.$

Example : calculation of information gain for an attribute of continuing value.

	70		85		90		95	
interval	<=	>	<=	>	<=	>	<=	>
CLASS1	2	0	2	0	2	0	2	0
CLASS2	0	3	1	2	2	1	3	0
Entropy	0	0	0.918	0	1	0	0.971	0
Info(S,T)	0		0.550		0.8		0.971	
Gain	0.971		0.421		0.171		0	

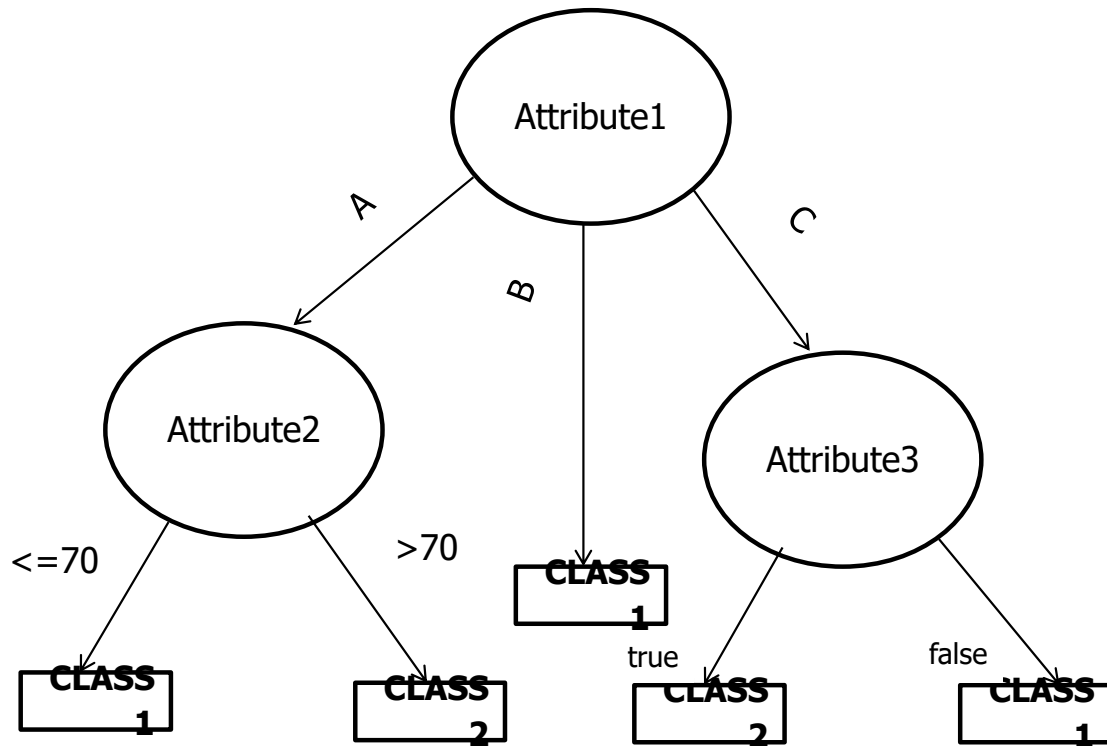
GAIN CALCULATION FOR THE ATTRIBUTE CONTINUOUS Attribute2 USING C4.5 ALGORITHM

- $\text{Entropy}(\leq 70) = 0$
- $\text{Entropy}(> 70) = 0$
- $\text{Info}(S, T) = (2/5) * 0 + (3/5) * 0 = 0$
- $\text{Gain} = 0.9710 - 0 = 0.971$ bits

Example: Choosing the Level 3 Node

- Gain (Attribute1, Attribute2) == 0.971 bits
- Gain (Attribute1, Attribute3) = 0.021 bits
- Attribute2 gives the highest gain of 0.971 bits.

decision tree



decision rules

```
If      Attribute1 = A
  Then
    If      Attribute2 <= 70
      Then
        Classification = CLASS1;
      Else
        Classification = CLASS2;
    Elseif Attribute1 = B
      Then
        Classification = CLASS1;
    Elseif Attribute1 = C
      Then
        If      Attribute3 = True
          Then
            Classification = CLASS2;
          Else
            Classification = CLASS1.
```

Data Mining



Bayes Classifier and Evaluation and Credibility

Bayes Classifier

- A probabilistic framework for solving classification problems

- Probability of an even A

$$P(A) = \frac{\#instance(A)}{\#the\ total\ of\ instances}$$

- Conditional Probability:

القانون يعني : ما احتمال حدوث C اذا اعطيت A

- Bayes theorem:

$$P(C | A) = \frac{P(A, C)}{P(A)}$$

$$P(A | C) = \frac{P(A, C)}{P(C)}$$

$$P(C | A) = \frac{P(A | C)P(C)}{P(A)}$$